

Clovis SFEIR

+33 6 95 83 95 34 · clovis.sfeir.cs@gmail.com · github.com/sfeirc · linkedin.com/in/clovis-sfeir-388936240 · cloooooo.com · French (Native) – English (C1)

Profile

IMT Atlantique engineering student specialising in systems security and low-level protocol engineering. Built five systems (C++20/Rust/Python) benchmarked against industry references or labeled ground-truth: an IEC-104 ICS intrusion detector achieving **100% attack recall** on real attack patterns (the exact protocol behind the 2016 Kyiv blackout), an HFT limit-order-book matching engine at **sub-microsecond P50 latency**, an EWMA/CUSUM statistical anomaly-detection pipeline (**0/10 false positives**), and an Ornstein-Uhlenbeck crude-spread trading engine.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*
Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France *Class rank: 1/85 | Software Engineering, Databases, Cybersecurity*
HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) — **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePlace (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

ICS Intrusion Detection (IEC 60870-5-104) *C++20, protocol parsing, explainable rules*

- Rule-based detector for IEC 60870-5-104 (no auth/encryption by protocol design) – the actual telecontrol protocol behind the 2016 Kyiv blackout – mapped to real attack patterns: control commands to unexpected breaker objects (the exact method used), sequence-number gaps (replay/desync), u-format/command floods.
- Parses APCL+ASDU at **39.5 ns (25.3 M msg/s)**; end-to-end parse+detect 337 ns; **100% recall, 0 false positives** on a labeled synthetic trace (100 records, 15 attacks across 5 rules); 153 assertions/37 tests, ASan+UBSan clean.

IIoT Predictive Maintenance *Rust, tokio, EWMA/CUSUM*

- Async telemetry pipeline (**tokio mpsc**) fuses EWMA + CUSUM control charts across 3 sensor channels to detect developing anomalies in real time – the same statistical alerting technique used to distinguish a real threat signal from background noise.
- Sensor-fusion tuning (2-of-3 channels + widened control limits) cut false positives from a 0.41–0.50 precision baseline to **0/10 across 10 seeds**; precision **1.0000**, recall 0.9642, F1 0.9818; **1.6 M samples/s**; 26 tests, 0 failures.

HFT Limit Order Book Matching Engine *C++20, cache-aware layout, intrusive data structures*

- Price-time-priority matching engine (limit/market/cancel) with intrusive doubly-linked price-level lists (**O(1) cancel** via pointer unlink) and pre-allocated object pooling to keep the hot loop free of **malloc/new**.
- Benchmark suite across three synthetic order-flow regimes: **2.3M+ msg/s** single-threaded throughput, **P50 0.6–1.0 μs** latency, zero hot-path allocations; unit tested, ASan+UBSan clean – memory-safety verified, not assumed.

Crude Oil Spread Trading Engine *C++20, stochastic processes, order book*

- Models refining margins (3:2:1 crack spread, generalized N:M:K) and mean-reverting inter-crude spreads via an Ornstein-Uhlenbeck process (Euler-Maruyama) with a rolling z-score signal, matched by a price-time-priority order book.
- OU stationary distribution verified against theory (500K-step sim); crack-spread formula checked to 10^{-9} ; 58 assertions/21 tests, ASan+UBSan clean.

Engineering Docs RAG *Python, BM25, information retrieval*

- From-scratch Okapi BM25 retrieval engine (chunking + ranking) for engineering documents; pluggable extractive/LLM generator interface, zero paid API calls in tests/CI/demo by design.
- Hand-derived BM25 scoring verified to 9 significant figures; measured **precision@1 0.90**, recall@5 1.00, **MRR 0.944** on a 40-question labeled set; 53 tests, zero runtime dependencies.

Work Experience

Infotel *Nantes, France*
AI Engineer Apprentice – Lab IA *Sep 2025 – Jun 2028*

- Built a Rust-based autonomous software-delivery harness that takes a complete specification through architecture synthesis (subsystems, interface contracts), epic/slice decomposition, and TDD-first generation to a tested, deployable application end-to-end; automated compile+test quality gates with iterative self-repair, independent code-review pass, and cross-slice conflict detection before delivery; 928 Rust tests.

Excelia Group *La Rochelle, France*
Software Developer Intern / AI Software Engineer Intern *Two internships: Jan–Mar 2025 & May–Jul 2024*

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

Security / Systems	ICS/SCADA protocol parsing (Modbus, IEC-104), explainable rule-based detection, ASan/UBSan memory-safety verification, cache-aware layout
Languages	C++20, Rust, Python, TypeScript, SQL
ML / Quant	EWMA/CUSUM change-detection, BM25/information retrieval, Ornstein-Uhlenbeck, Kalman filter
Infrastructure	Docker, tokio async, Prometheus/Grafana, GitHub Actions CI, GCP, Redis
Frameworks	Flask, Node.js/Express, React/Next.js, REST/gRPC